



The Path to Becoming a DevOps Engineer

Career choices can often baffle newbies. Knowledge is the key in making a decision. Here's a guide to career options in the DevOps engineering field. This article will guide new entrants to the IT industry on the many roles that DevOps engineers can assume, the job requirements and the certifications necessary to obtain those positions in the IT industry. To be forewarned is to be forearmed.

In the 21st century, technology has evolved to meet the challenges faced in IT management across various segments. The technological advances and changes have been sweeping, rapid, and encompass multiple fields like the Agile methodology, DevOps, Big Data, cloud computing and machine learning. Prior to DevOps, software development faced countless challenges like changes in the system, deployments that were fraught with risks, and the lack of consistency across platforms. Development teams struggled to overcome these hurdles.

DevOps (development plus operations) has now been taken to the central stage of the software development life cycle (SDLC). It offers process frameworks mixed with open source tools to integrate all phases of the application's life cycle, and ensures that everything functions as a cohesive unit. It aims to align and automate the process across the phases of development, testing, deployment and support. It includes the best methodologies and practices such as code repositories, build automation and continuous deployment, among others.

DevOps can be defined as, "A set of practices intended to reduce the time between committing a change to a system and the change being placed into normal production, while ensuring high quality."

It can also be defined as, "An evolution of the ideas of Agile software development and lean manufacturing, applied to the end-to-end value chain in IT, which allows businesses to achieve more with modern information technologies, due to cultural, organisational and technical changes."

According to the Second Annual GitLab Global Developer Survey conducted in March 2018, "Thirty-five per cent of the respondents have already established the DevOps culture and a mere 23 per cent describe their development method as DevOps." The survey adds, "Sixty-five per cent of the respondents believe that the DevOps workflow saves them time during the development process and 29 per cent plan to invest in DevOps in 2018."

The worldwide survey indicates a steady increase in DevOps adoption by enterprises from varied sectors and regions. As a set of cultural philosophies, practices and tools,